'Assignment-04'

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Sec: 10

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Question - 01

(a)

i) IN AX, 2A40H

This is Direct (fixed) port addressing. However, this is an invalid addressing because, in 8086, the direct I/O addressing requires a 8 bit port number in the instruction itself. The given port number is 16 bit which is wrong.

ii) MOV DX, 2A40H
IN AX, DX

This is indirect (variable) port addressing where the 16 bit port number is loaded into DX register first then accessed.

iii) MOV AX, [2A40H]

This is Direct memory addressing but in I/O accessing context used as memory-mapped I/O technique. The offset (2A40H) is the port number accessed relative to DS.

(b)

The first instruction (IN AX, 2A40H) is invalid.

Direct (fixed) port addressing requires an 8-bit port number to be provided so it can address port (00-FFH). The given 2A40H port number is a 16-bit number which doesn't fit the p8 field. So, this is invalid.

(c)

Yes, the I/O instruction technique showcased in the 3rd instruction can access a greater number of ports compared to the one in 2nd instruction.

The 2nd instruction uses isolated I/O technique where IN or OUT opcode is used to access I/O devices. In this, (AD0-AD15), a total 16 pins are used to access the I/O devices address. So, using this, the number of addressable port is,

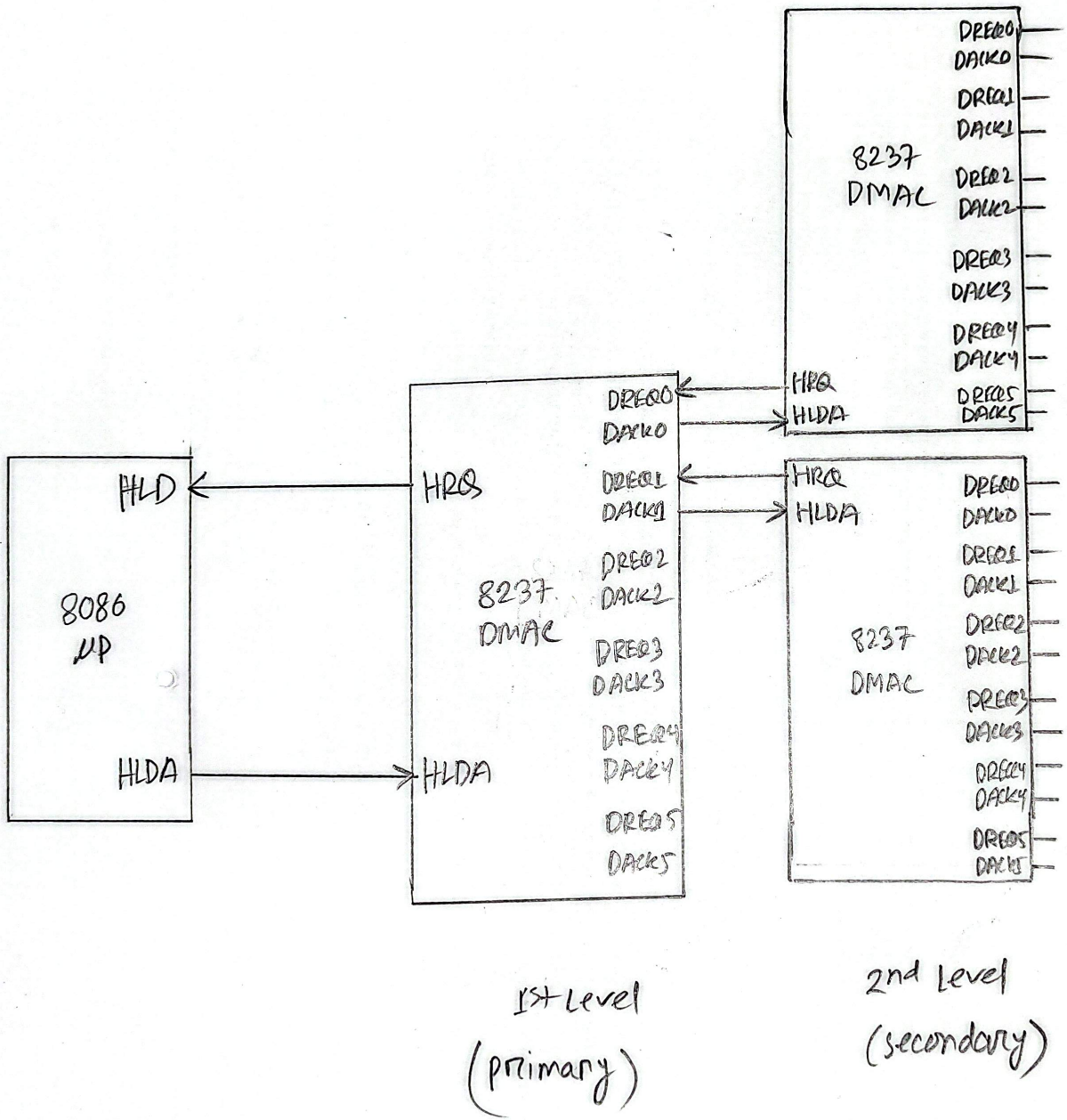
$$2^{16} = 65,536$$

However, the 3rd instruction uses memory mapped I/O addressing technique where a part of the memory is used to map I/O devices. In 8086, there is a 20-bit address bus which provides a physical address space of 2^{20} or,

$$2^{20} = 1048576$$

As we can see, the 3rd instruction has a greater number of addressable ports compared to 2nd instruction. So, the statement is true.

(d) (i)



1st Level
(primary)

2nd Level
(secondary)

(ii)

The sequence of operation during DMA-based data transfer is given below:

1) The microprocessor's CPU invokes a read/write command to the peripherals.

2) The peripheral device when ready sends a DREQ signal to one of the secondary DMA controller.

3) The secondary DMA controller request primary controller through HRQ that is connected to DREQ pin.

4) The primary controller then request microprocessor to get the control over the system bus through HRQ pin connected to HLD of microprocessor's.

5) Then microprocessor completes its current machine cycle and acknowledge the primary controller request through HLDA pin connected to primary controller's HLDA.

6) The primary controller gets the access over the system bus and acknowledge secondary controller request through DACK connected to requesting secondary controller's HLDA.

7) The secondary controller recognizes that it has the control now and acknowledge the requesting peripheral

device through DACK pin.

8) The secondary DMA controller here performs all the data transfer and generates necessary control signals, memory address.

9) Data transfer happens and DMA controller sends signal to CPU dropping their HRQ signal and disabling DACK signal. CPU disables HLDA and takes back control of the system bus.

(iii)

During a write cycle:

i) HRQ = 1 ; controller is requesting control of system bus from the CPU.

ii) HLDA = 1 ; CPU has granted bus control to the controller

iii) DREQ = 1 ; Peripheral device requesting service.

iv) DACK = 1 ; Controller acknowledge the device's request.

v) $\overline{IOR}$ = 0 ; Allows reading data from I/O device.

vi) $\overline{IOW}$ = 1 ; Currently not writing to I/O.

vii) $\overline{MEMW}$ = 0 ; Data is written into memory.

viii) $\overline{MEMR}$ = 1 ; Currently not reading memory.

(e)

(i)

given DMA6 controller has 6 channels. If one secondary controller is attached, the number of channel becomes $(6+5) = 11 > 8$.

So, 1 secondary DMA6 controller is minimum to adopt utility of 8 channels.

(ii)

If DREQ5 and DACK5 are faulty, the number of channels are 5 in DMA6 that is working. Again, if one DMA6 controller is used as secondary controller, the number of channels become $(5+4) = 9 > 8$.

Still 1 minimum DMA6 controller is enough to adopt utility of 8 channels.

(iii)

If DREQ1 and DACK5 pins are faulty, the number of channels working in DMA6 becomes 4.

Now, if one secondary controller is used, the number of

channels become $(4+3) = 7 < 8$.

So, one secondary controller is not enough. If two are used then, the number of channels become $(4+4+2) = 10 > 8$

So, two secondary DMA6 controller is minimum for adopting 8 channels. Hence, the number changes.

Question-2

(a)

The starting address for INT 27:

$27 = 0001BH$

$\therefore$ starting address $= (0001B \times 4)H$

$= 0006CH$

Now, from starting address we get CS:IP of the instruction.

$0006CH = 00H \rightarrow (IP_L)$

$0006DH = 03H \rightarrow (IPH)$

$0006EH = 00H \rightarrow (CS_L)$

$0006FH = 05H \rightarrow (CSH)$

$\therefore$ CS : IP = 0500 : 0300

and address $= (0500 \times 10) + 0300 = 05300H$

So, the first two byte instruction fetched:

$$05300H = 89H$$
$$05301H = 8BH$$

∴ The first 2-byte hex instruction is 898BH.

(b)

The INT 27 is non-maskable interrupt.

INT 27 is a software interrupt explicitly invoked by an instruction. The IF flag has no influence on software type interrupts and so it is non-maskable.

(c)

The given MOV DH, [SI + XXXXh] is a 4-byte instruction.

So, the total size of ISR:

$$(25 \times 4)\blacksquare \text{ bytes} = 100 \text{ bytes}$$

Now, the starting address of ISR is 05300H and the IRET is located immediately after 25 MOV instructions.

So, physical address of the IRET instruction is:

100 = 00064H.

∴ IRET physical address = (05300 + 00064H)

= 05364H

∴ IRET instruction physical address is 05364H.

(d)

Given, IR6 and IR7 are dead on the PICs. So, each PIC has 6 usable IR lines.

Also, CAS2 is said to be stuck at 0. Means only 000 to 011 can be addressed. so, only 4 slaves PIC can be cascaded.

Now, if Master's IR0 - IR3 has cascaded PIC, the number of interrupt:

(4 slave x 6 usable IR) = 24 interrupts

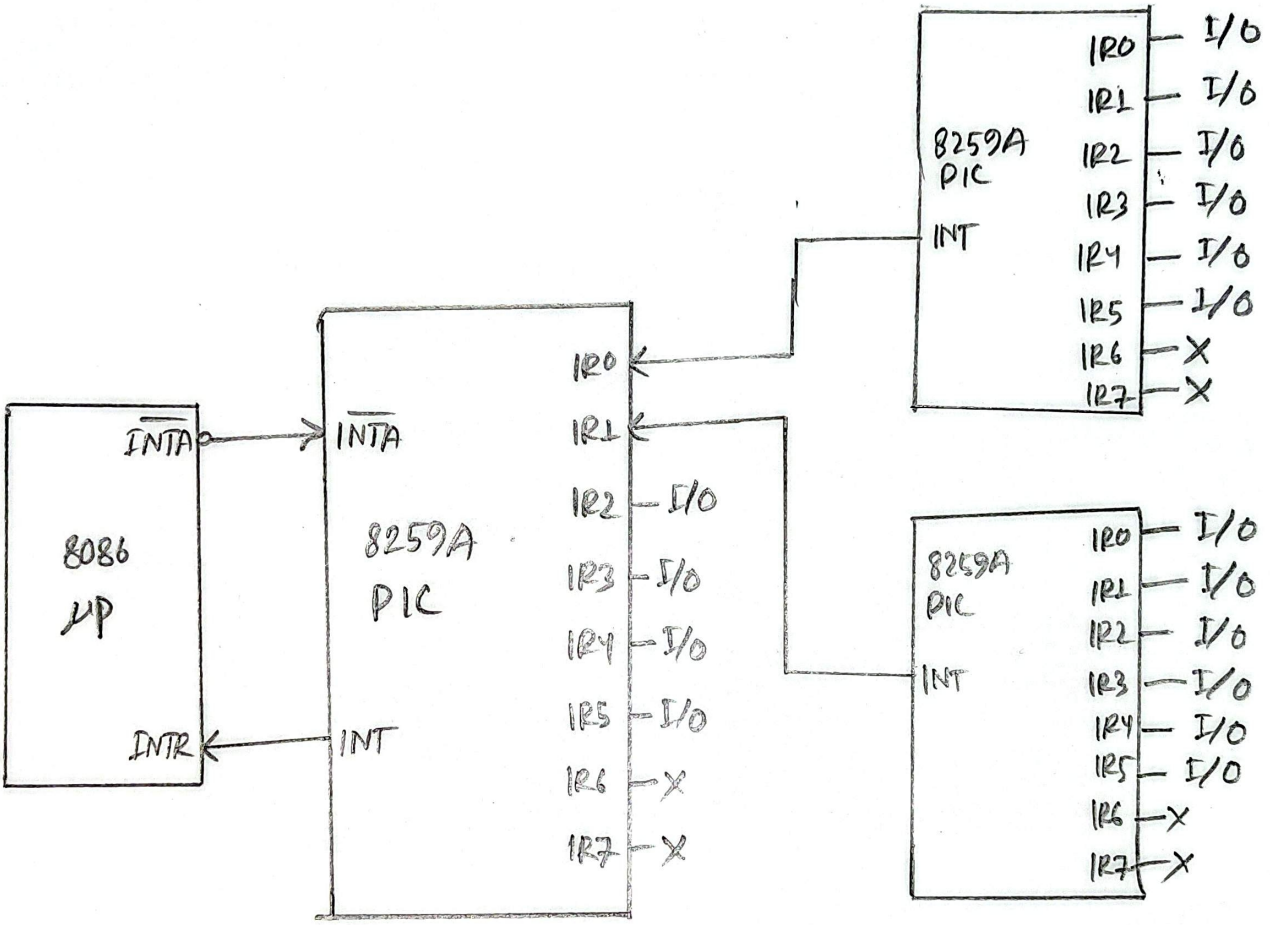
Master PIC has IR4 and IR5 working but not cascaded to any slave PIC.

So, total maximum number of interrupt is

= (24+2) = 26 interrupts.

(e)

If the same faulty PIC is used where IR6 and IR7 is dead pins and CAS2 stuck at 0, then to address 16 interrupts, 2 slave PIC is required minimum.



As we can see two slave PIC are connected to master PIC giving a total of 16 interrupt.

Question-3

(a)

OUT AL, 04H

This is an invalid instruction. In 8086 the destination operand requires a port number. The given instruction has a register. The valid form would be,

OUT 04H, AL

(c)

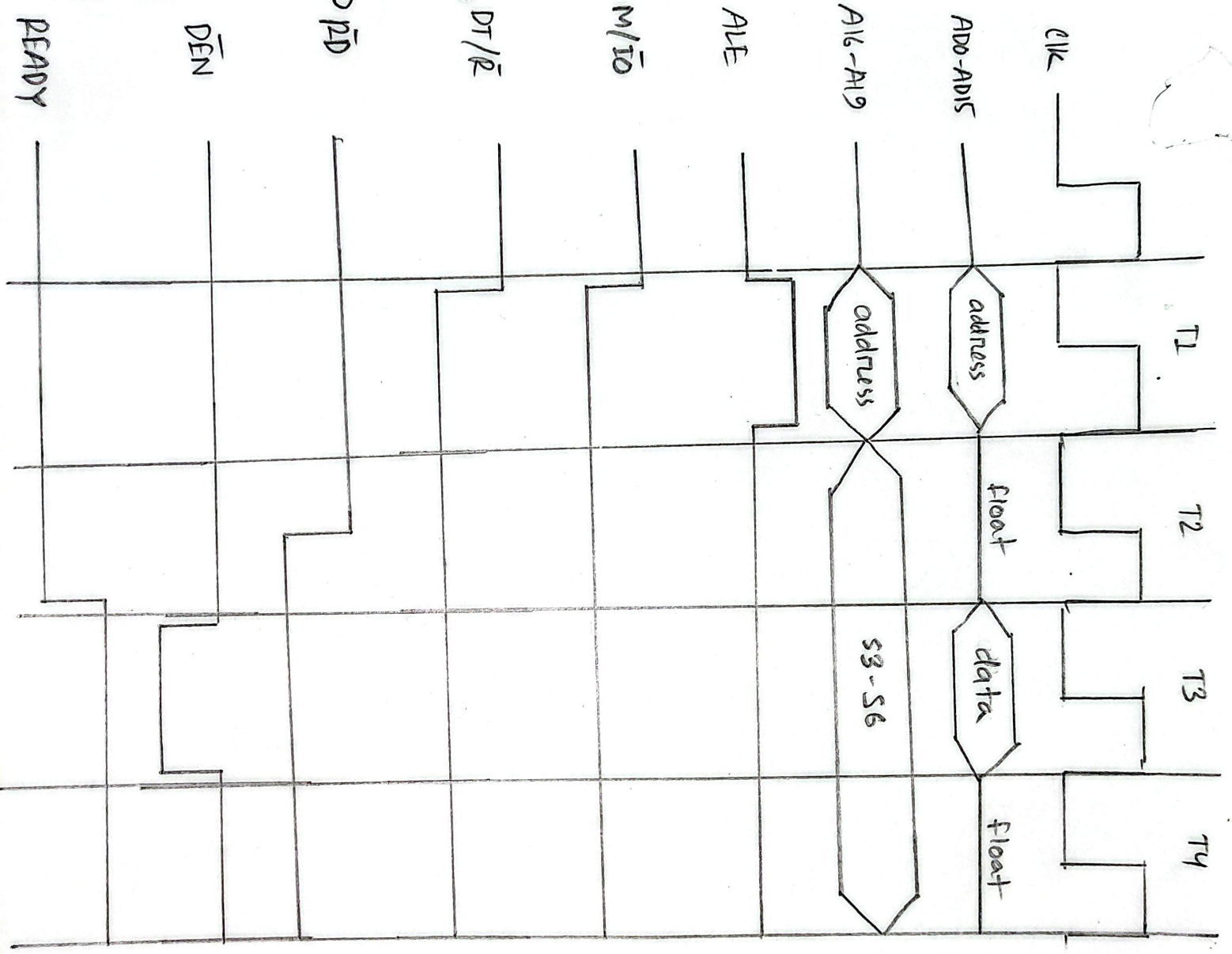
OUT AX, 02H

This is an invalid instruction. In 8086, the destination operand requires a port number for OUT instruction. The given instruction has a register. The valid form would be,

OUT 02H, AX

(b)

IN AL, 05H requires one bus cycle. It receives 8-bit data from an add port address. The processor fetches the data in a single cycle over (D8-D15).



(d)

IN AX, 03H requires 2 bus cycles. Since the given odd port address is unaligned one bus cycle is for reading odd byte another to read high byte.

